

## Industrial Internship Report on " Prediction of Agriculture Crop Production in India"

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### *Executive Summary*

This report presents the work completed during the Industrial Internship organized by upskill Campus and The IoT Academy in collaboration with UniConverge Technologies Pvt Ltd (UCT).

During this internship, I worked on the project "**Prediction of Agriculture Crop Production in India**", which focuses on predicting crop yield using Machine Learning techniques. The project involved data preprocessing, feature engineering, model training, and performance evaluation.

A Random Forest Regression model was developed using agricultural data related to crop type, state, cultivation cost, and production cost. The model achieved an **R<sup>2</sup> score of 0.9463**, demonstrating good predictive performance.

This internship provided practical exposure to Data Science and Machine Learning concepts and enhanced my analytical, problem-solving, and project development skills.

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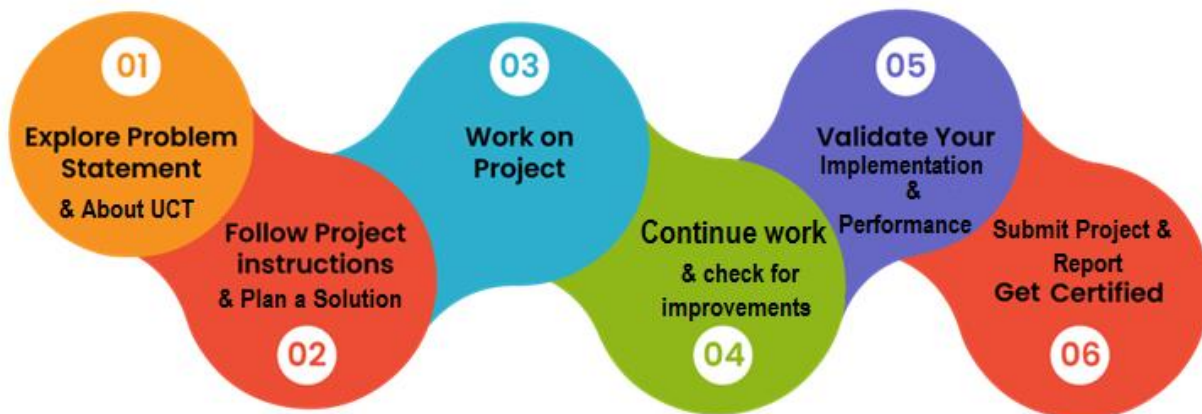
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## 1 Preface

This internship was conducted under **upskill Campus** and **The IoT Academy** in collaboration with **UniConverge Technologies Pvt Ltd (UCT)** to provide practical exposure to Data Science and Machine Learning concepts.

During the six-week internship, I worked on the project "**Prediction of Agriculture Crop Production in India**". The project involved analyzing agricultural data, preprocessing datasets, developing a Machine Learning model, and evaluating its performance for crop yield prediction.

This internship provided me with valuable experience in Python programming, data analysis, machine learning model development, and data visualization. It helped me understand how technology can be applied to solve real-world problems in the agriculture sector.



### My Learnings and Overall Experience

I gained hands-on experience in data preprocessing, feature engineering, model training, and performance evaluation. The internship enhanced my analytical thinking and problem-solving skills.

### Acknowledgement

I would like to thank **UCT**, **upskill Campus**, **The IoT Academy**, and **KIIT University** for providing this learning opportunity and supporting me throughout the internship.

### Message to Juniors and Peers

I encourage students to participate in internships and project-based learning programs as they provide practical knowledge and industry exposure beyond classroom learning.

## 2 Introduction

### 2.1 About UniConverge Technologies Pvt Ltd

A company established in 2013 and working in Digital Transformation domain and providing Industrial solutions with prime focus on sustainability and RoI.

For developing its products and solutions it is leveraging various **Cutting Edge Technologies** e.g. **Internet of Things (IoT), Cyber Security, Cloud computing (AWS, Azure), Machine Learning, Communication Technologies (4G/5G/LoRaWAN), Java Full Stack, Python, Front end** etc.



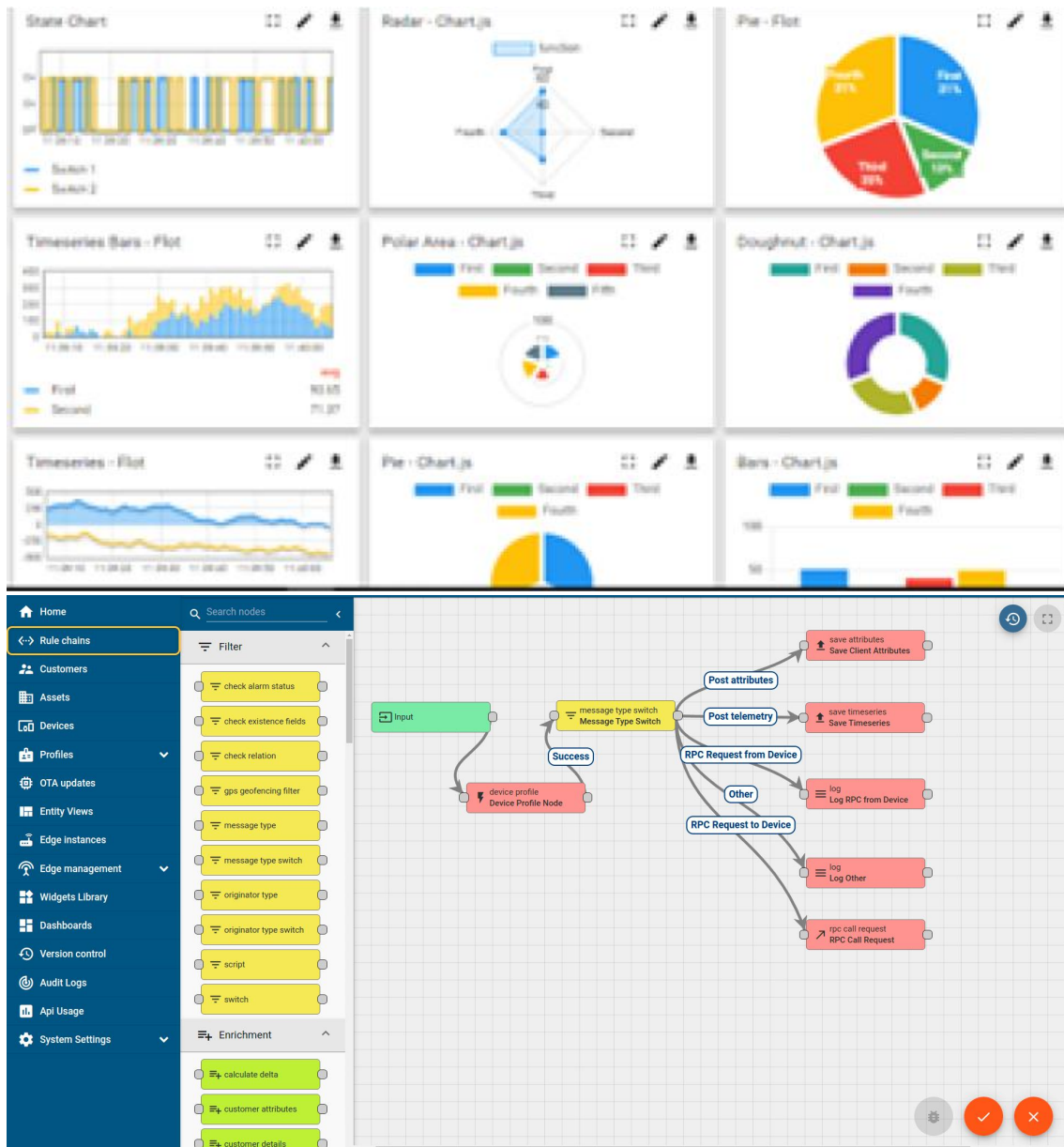
#### i. UCT IoT Platform ( )

**UCT Insight** is an IOT platform designed for quick deployment of IOT applications on the same time providing valuable “insight” for your process/business. It has been built in Java for backend and ReactJS for Front end. It has support for MySQL and various NoSql Databases.

- It enables device connectivity via industry standard IoT protocols - MQTT, CoAP, HTTP, Modbus TCP, OPC UA
- It supports both cloud and on-premises deployments.

It has features to

- Build Your own dashboard
- Analytics and Reporting
- Alert and Notification
- Integration with third party application(Power BI, SAP, ERP)
- Rule Engine



## FACTORY WATCH

### ii. Smart Factory Platform ( )

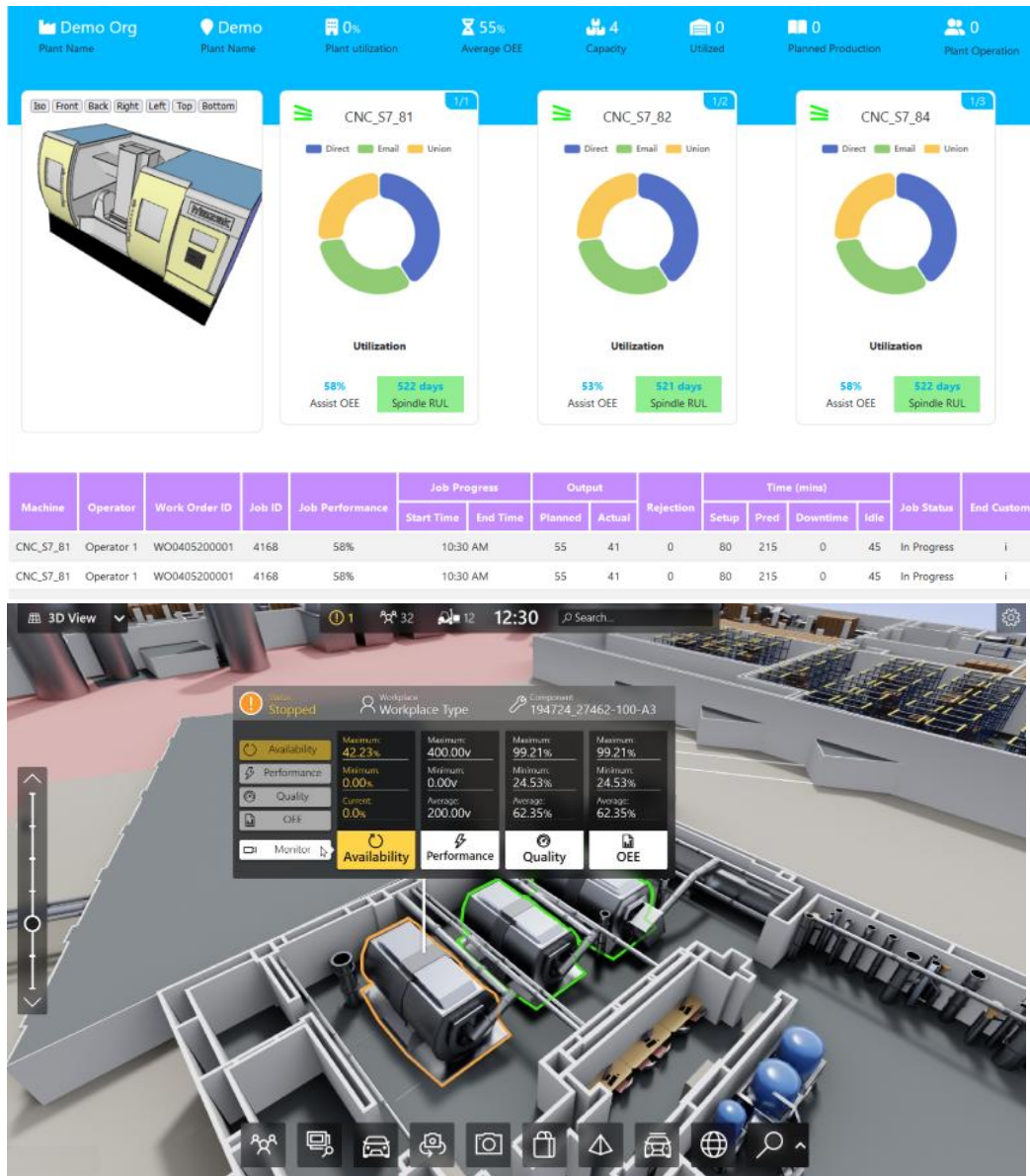
Factory watch is a platform for smart factory needs.

It provides Users/ Factory

- with a scalable solution for their Production and asset monitoring
- OEE and predictive maintenance solution scaling up to digital twin for your assets.
- to unleash the true potential of the data that their machines are generating and helps to identify the KPIs and also improve them.
- A modular architecture that allows users to choose the service that they want to start and then can scale to more complex solutions as per their demands.

Its unique SaaS model helps users to save time, cost and money.





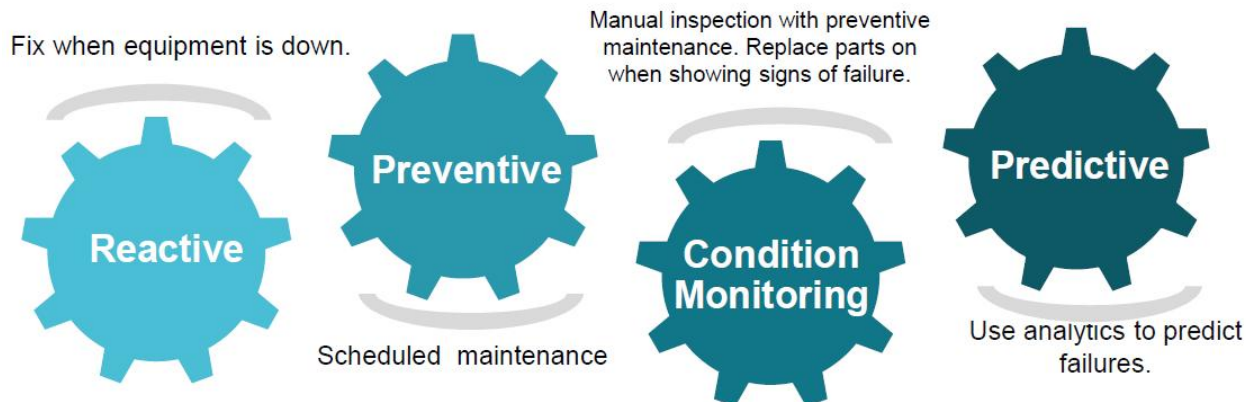


### iii. based Solution

UCT is one of the early adopters of LoRAWAN technology and providing solution in Agritech, Smart cities, Industrial Monitoring, Smart Street Light, Smart Water/ Gas/ Electricity metering solutions etc.

### iv. Predictive Maintenance

UCT is providing Industrial Machine health monitoring and Predictive maintenance solution leveraging Embedded system, Industrial IoT and Machine Learning Technologies by finding Remaining useful life time of various Machines used in production process.

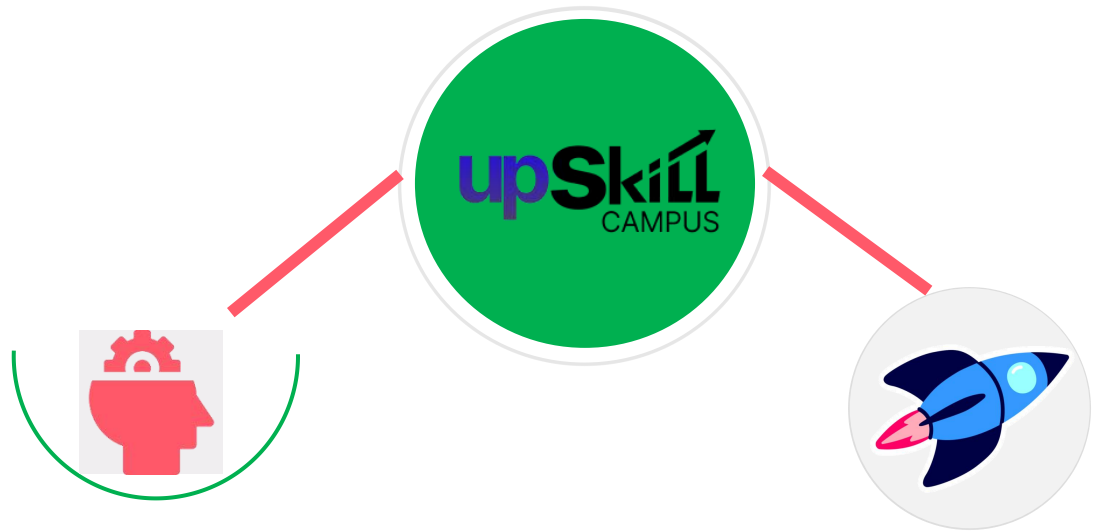


## 2.2 About upskill Campus (USC)

upskill Campus along with The IoT Academy and in association with Uniconverge technologies has facilitated the smooth execution of the complete internship process.

USC is a career development platform that delivers **personalized executive coaching** in a more affordable, scalable and measurable way.

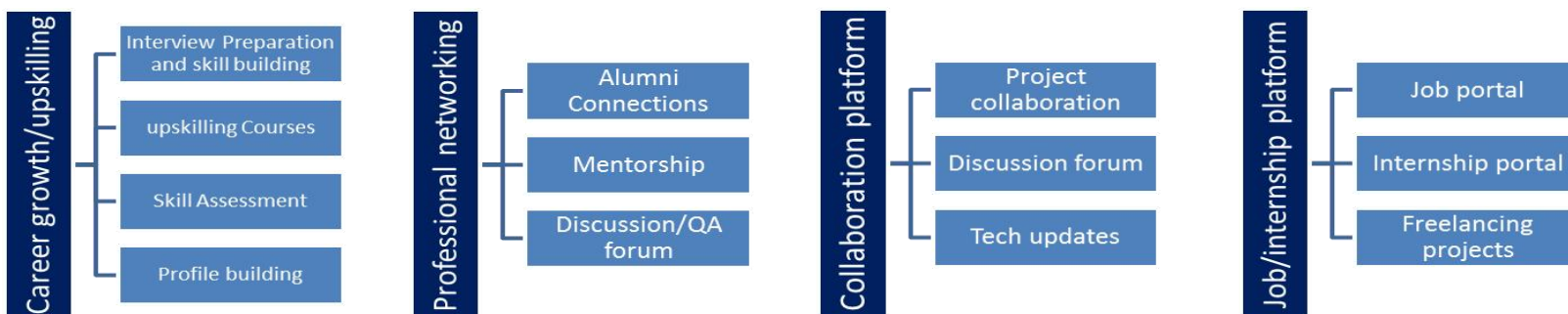




Seeing need of upskilling in self paced manner along-with additional support services e.g. Internship, projects, interaction with Industry experts, Career growth Services

upSkill Campus aiming to upskill 1 million learners in next 5 year

<https://www.upskillcampus.com/>



## 2.3 The IoT Academy

The IoT academy is EdTech Division of UCT that is running long executive certification programs in collaboration with EICT Academy, IITK, IITR and IITG in multiple domains.

## 2.4 Objectives of this Internship program

The objective for this internship program was to

- get practical experience of working in the industry.
- to solve real world problems.
- to have improved job prospects.
- to have Improved understanding of our field and its applications.
- to have Personal growth like better communication and problem solving.

## 2.5 Reference

[1] Scikit-Learn Documentation – <https://scikit-learn.org>

[2] Pandas Documentation – <https://pandas.pydata.org>

[3] Matplotlib Documentation – <https://matplotlib.org>

## 2.6 Glossary

| Terms                        | Acronym        |
|------------------------------|----------------|
| Machine Learning             | ML             |
| Random Forest                | RF             |
| Mean Absolute Error          | MAE            |
| Root Mean Square Error       | RMSE           |
| Exploratory Data Analysis    | EDA            |
| Coefficient of Determination | R <sup>2</sup> |

### 3 Problem Statement

Traffic congestion has become a major challenge in modern cities due to the increasing number of vehicles and limited road infrastructure. Accurate traffic prediction can help city authorities optimize traffic management, reduce congestion, improve travel time estimation, and enhance urban mobility.

The objective of this project is to develop a Machine Learning model capable of predicting traffic volume at different junctions based on historical traffic data. The dataset contains information such as date, time, junction details, and vehicle counts.

By analyzing traffic patterns and temporal features, the proposed solution aims to provide reliable traffic predictions that can support smart city planning and intelligent transportation systems.

## 4 Existing and Proposed solution

### Existing Solution

Traditional traffic monitoring systems rely on manual observation, fixed scheduling, and basic statistical analysis. These approaches often fail to capture dynamic traffic patterns and are unable to provide accurate future traffic forecasts.

### Proposed Solution

The proposed system utilizes Machine Learning techniques to predict vehicle traffic at road junctions. Historical traffic data is preprocessed and transformed into meaningful features such as year, month, day, hour, day of week, and weekend indicators. A Random Forest Regression model is trained on these features to predict future vehicle counts.

### Value Addition

- Automated traffic prediction.
- Improved traffic management and planning.
- Reduced congestion through data-driven decisions.
- Scalable solution for smart city applications.
- Can be integrated with real-time traffic monitoring systems.

### 4.1 Code submission (Github link):

Code link: -> [Link](#)

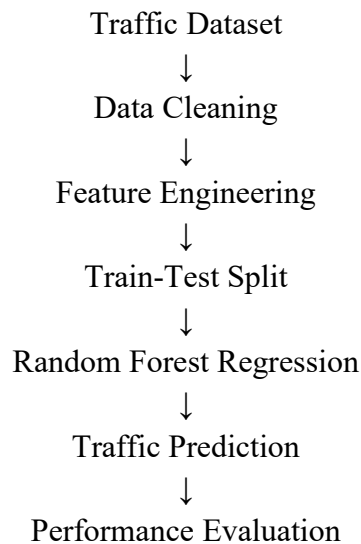
### 4.2 Report submission (Github link) :

Report link:-> [link](#)

## 5 Proposed Design/ Model

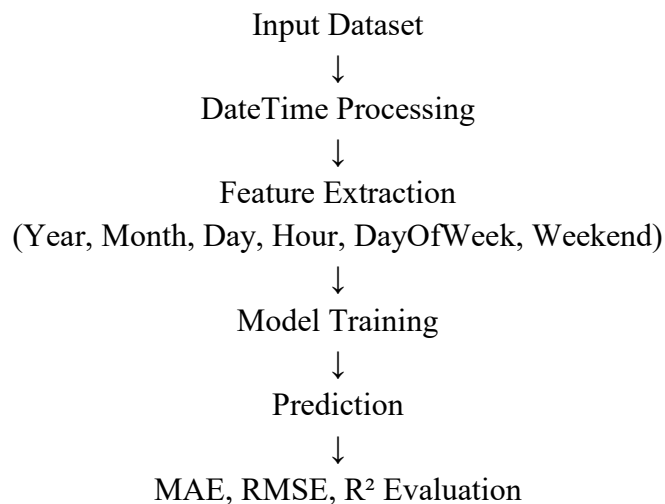
The proposed system follows a Machine Learning workflow for traffic prediction. Historical traffic data is collected and preprocessed before extracting temporal features. The processed data is used to train a Random Forest Regression model, which predicts vehicle counts at various junctions.

- **5.1 High Level Diagram**



**Figure 1: High Level Diagram of Smart City Traffic Prediction System**

- **5.2 Low Level Diagram**





↓  
Visualization

**Figure 2: Low Level Diagram of System Workflow**

- **5.3 Interfaces**

Dataset Interface:

- Historical traffic dataset containing DateTime, Junction, Vehicle Count, and ID.

Processing Interface:

- Data preprocessing using Pandas.
- Feature engineering from DateTime.
- Model training using Random Forest Regressor.

Output Interface:

- Predicted traffic volume.
- Performance metrics (MAE, RMSE,  $R^2$ ).
- Traffic visualization graphs.

## 6 Performance Test

The Smart City Traffic Prediction model was evaluated using standard regression metrics to assess its ability to accurately predict vehicle traffic volume at various junctions. Historical traffic data was analyzed and transformed into time-based features to improve prediction accuracy.

The primary constraints considered during model development were:

- Variations in traffic flow across different hours of the day.
- Differences in traffic volume among junctions.
- Large dataset size requiring efficient processing.
- Requirement for accurate predictions for smart city planning.

To address these constraints, feature engineering techniques were applied to extract temporal information such as year, month, day, hour, day of week, and weekend indicators. A Random Forest Regression model was then trained and evaluated.

### 6.1 Test Plan / Test Cases

| Test Case           | Description                                                 | Expected Result                    |
|---------------------|-------------------------------------------------------------|------------------------------------|
| Data Loading        | Load traffic dataset                                        | Dataset loaded successfully        |
| Feature Engineering | Extract DateTime features                                   | Features generated correctly       |
| Model Training      | Train Random Forest model                                   | Model trained successfully         |
| Prediction          | Generate traffic predictions                                | Predicted vehicle counts generated |
| Evaluation          | Calculate performance metrics MAE, RMSE, and $R^2$ obtained |                                    |

### 6.2 Test Procedure

- Load and inspect the traffic dataset.
- Convert DateTime column into datetime format.
- Extract Year, Month, Day, Hour, DayOfWeek, and Weekend features.

- Split data into training and testing sets.
- Train the Random Forest Regression model.
- Generate predictions on test data.
- Evaluate model performance using MAE, RMSE, and  $R^2$  metrics.
- Visualize the traffic patterns using graphs.

### 6.3 Performance Outcome

#### Model Performance

| Metric | Value |
|--------|-------|
|--------|-------|

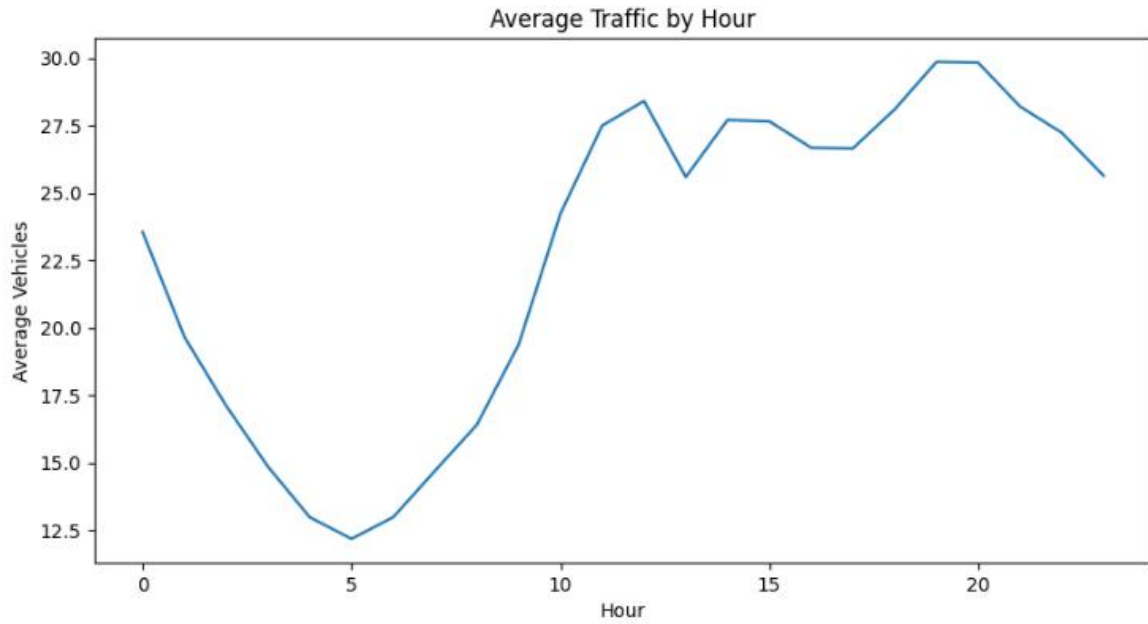
|     |      |
|-----|------|
| MAE | 2.39 |
|-----|------|

|      |      |
|------|------|
| RMSE | 3.55 |
|------|------|

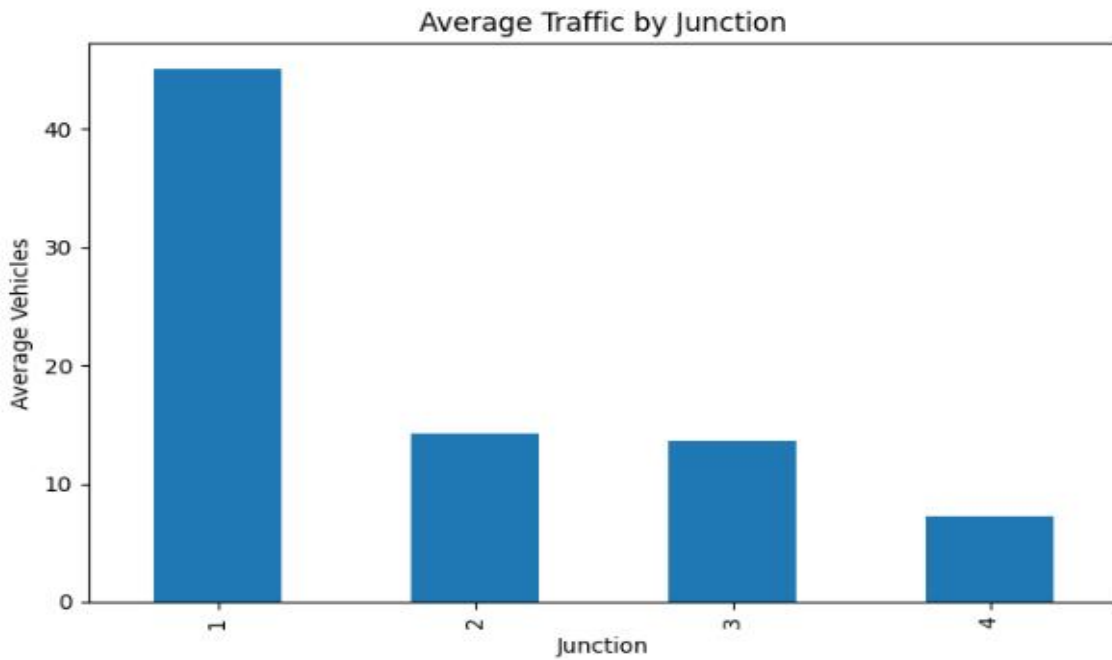
|             |       |
|-------------|-------|
| $R^2$ Score | 0.969 |
|-------------|-------|

#### Result Analysis

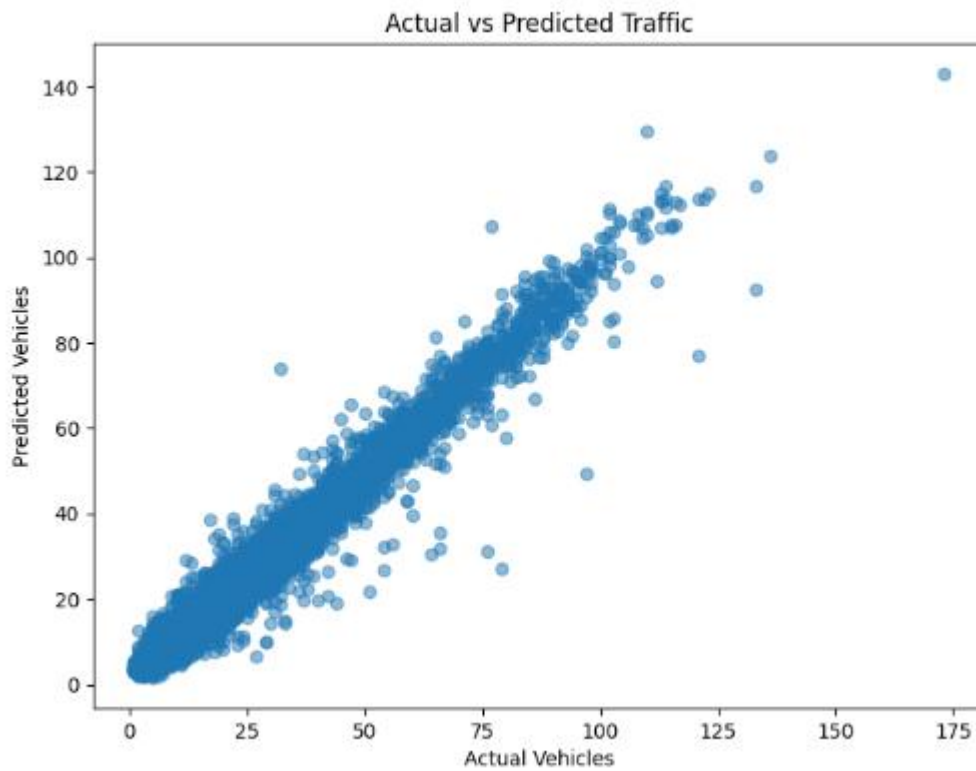
The model achieved an  $R^2$  score of 0.969, indicating excellent predictive capability. The low MAE and RMSE values demonstrate that the model accurately captures traffic patterns and vehicle flow variations across different junctions and time periods.



**Figure 3: Average Traffic by Hour**



**Figure 4: Average Traffic by Junction**



**Figure 5: Actual vs Predicted Yield**

## 7 My learnings

During this internship, I gained practical exposure to the complete Machine Learning project lifecycle, including data collection, preprocessing, feature engineering, model development, evaluation, and result visualization.

I improved my skills in:

- Python programming for data analysis.
- Data preprocessing using Pandas.
- Data visualization using Matplotlib.
- Machine Learning model development using Scikit-Learn.
- Feature encoding and model evaluation techniques.
- Applying Machine Learning to solve real-world agricultural problems.

Working on the project "**Smart City Traffic Prediction**" helped me understand how data-driven solutions can be used to support decision-making in the agriculture sector. It also strengthened my analytical thinking, problem-solving abilities, and project development skills.

This internship has increased my confidence in building end-to-end Machine Learning projects and has provided valuable experience that will contribute to my future academic and professional growth.

## 8 Future work scope

- Integrate real-time traffic sensor data for live traffic prediction.
- Incorporate weather and event information to improve forecasting accuracy.
- Compare advanced machine learning algorithms such as XGBoost and Gradient Boosting.
- Develop a web-based dashboard for traffic monitoring and visualization.
- Extend the system to provide route optimization and congestion management recommendations.
- Integrate the model with smart city infrastructure and intelligent transportation systems.